The first part of this chapter is a high-level exposition on information theory, entropy, and ways to maximize gain in the amount of information we receive upon performing an action. This will be done by reinterpreting the familiar example of binary search. Armed with the fundamentals of information theory, we then move on to discussing BayesOpt policies that maximize information about the global optimum of an objective function. These policies are the result of applying information theory to the task of BayesOpt. As always, we will also learn how to implement these policies in Python.

By the end of the chapter, you will gain a working understanding of what information theory is, how entropy as a measure of uncertainty is quantified, and how it is translated to BayesOpt. This chapter adds another policy to our BayesOpt toolkit and concludes the second part of the book on BayesOpt policies.

6.1 Measuring knowledge with information theory

Information theory is a subfield of mathematics that studies how to best represent, quantify, and reason about information in a principled, mathematical manner. In this section, we will get introduced to information theory on a high level and discuss how it is related to decision making under uncertainty. We will do this by re-examining the idea behind binary search, a popular algorithm in computer science, from the perspective of information theory. This discussion will subsequently allow us to connect information theory to Bayesian optimization and motivate an information-theoretic policy for optimization.

6.1.1 Measuring uncertainty with entropy

Information theory is especially prevalent in computer science, where digital information is represented as bits (zeros and ones). You might remember doing calculations about how many bits are required to represent a given number of integers—for example, a single bit is enough to represent two numbers: 0 and 1, while five bits are necessary represent thirty two (two raised to the fifth power) different numbers—these calculations are one of the things that information theory concerns itself with.

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